- If you are importing a location from SAP Transportation Management (TM), set the **Source System** to the business system name of your TM system and set the **External ID** to the location ID in TM.

  For locations that have its object type set to "Business Partner", you can find the business system name of your TM system and the location ID in the SOAP request message sent to LBN. In this request, the *MessageHeader.SenderBusinessSystemID* attribute specifies the business system name and the internal ID of the sender (the *MessageHeader.SenderParty.InternalID* attribute) specifies the location ID.

  Alternatively, you can run transaction code *SLDCHECK* in TM to view the business system name.

- **Logical System**: You can also use the **Logical System** field to identify the system from which the location master data is replicated.

  This is a mandatory field if you want to use the location in SAP Business Network Global Track and Trace. The logical system ID you enter here must be the same one that you used when configuring the replication of location master data via the Data Replication Framework. For more information, see Replicating Locations from SAP S/4HANA via Data Replication Framework (DRF). Make sure you enter the logical system ID correctly so that it matches the source system of the location. Otherwise, there can be duplicate records in your location master data.

- (Optional) **Address**:
  - **Country/Region**
  - **State**
  - **City**
  - **District**
  - **Street/House Number**
  - **Zip Code**
- (Optional) **Geographical Coordinates And Map**:
  - **Longitude**: Longitude coordinate of the location
  - **Latitude**: Latitude coordinate of the location

  ### Note

  You can enter the coordinates in the **Longitude** and **Latitude** fields manually. If you want the system to verify your entries, set **Calibrate coordinates when saving** to **Yes** before you save your changes. You can also choose **Generate Longitude and Latitude Coordinates** to create the coordinates according to the address information you've entered.

  By default, **Calibrate coordinates when saving** is set to **Apply global settings**, meaning that how the system handles coordinates for the location record you are working on is determined by the option you choose in the **Generate geo-coordinates automatically** setting, which can be found in the **Settings** dialogue box. If you leave **Generate geo-coordinates automatically** with the default option, the system generates geo-coordinates for locations when both the latitude and longitude coordinates are missing. For more information, see Global settings for Geo-Coordinates.

- (Optional) **Time Zone**
- (Optional) **Alternative Location ID**:
  - **Type**: Type of the alternative ID.
  - **ID**: Alternative ID in the specified type.

- (Optional) **Contact Information**: Contact information related to the location.
- (Optional) **Additional Information**: Additional information about the location. For more detail, see Define Additional Information.

3. Choose **Save**.

Result

The newly-created location master data can be consumed by other apps in SAP Business Network Global Track and Trace.

Editing a Location

To edit a location, proceed as follows:

1. Use the following fields to filter locations:
   - Location ID
   - Description
   - Location Type
   - Country/Region
   - City
2. Click the location from the filter results to open the details page.
3. Choose **Edit** on the upper-right corner of the details page.

System Connections (Data Contributor)

Use the **System Connections** app for data contributors to complete the implementation of SAP Business Network for Logistics APIs.

By implementing the tracking APIs published by SAP Business Network Global Track and Trace, your organization can receive and respond to tracking requests directly from your system.

To complete the API implementations, you – as an administrator – also need to perform the following steps using the **System Connections** app:

- Specify the authentication details and base URLs for your system's endpoints.
- Assign authentication details and a base URL to each API operation that you've implemented.
- Enable inbound communication to SAP Business Network for Logistics.

The app has the following areas:

- Outbound Connections

  This area lists the outbound-from-network API services to which you can assign authentication details and a base URL. In addition, you can see the active or inactive status of the operations.

- Manage Details

This area enables you to manage (create, edit and delete) your base URLs (communication details) and your authentication details (OAuth 2.0 and basic credentials).

- Connection Settings

  This area enables you to call LBN endpoints for inbound and outbound connections.

For more information about the APIs published by SAP Business Network Global Track and Trace, go to https://api.sap.com/package/LbnGlobalTrackAndTrace?section=Overview.

Specifying the Authentication Details and Base URLs for Your System's Endpoints

Configure communication and authentication details for your system's API or EDI endpoints by specifying base URLs, protocols (HTTPS/SFTP/Email), and authentication methods (Basic, API Key, OAuth 2.0) to enable secure integration with SAP Business Network for Logistics.

Procedure

1. In the **System Connections** app, click **Manage Details**.
2. In the **Communication Details** section, click **Add** and complete the dialog.

   You can choose between the following options:

   - **HTTPS** (for API)
   - **SFTP** (for EDI)
   - **Email** (for EDI)
3. In the **Authentication Details** section, click **Add** and complete the dialog.

   When creating API communication details, you need to enter the following information:

Attribute	Description
Name	An identifier for the details
Base URL	The endpoint URL for the communication

   The API call made to your system uses the HTTPS protocol.

   When creating EDI communication details, you need to enter the following information:

Attribute	Description
Name	Enter a name to identify the authentication details in this app.
Address	Enter the host name or IP address of the SFTP server and an optional port, for example, `my.host.org:22`. You can configure this parameter by entering a dynamic expression such as `${property.property_name}` or `${header.header_name}`.
Message Format	You can choose EDIFACT or ANSI X12 from the drop-down list.

Attribute	Description
Message Type	For EDIFACT, you can choose between the following options: - IFTMIN - All Inbound Messages For ANSI X12, you can choose between the following options: - 204 - 213 - 300 - 303 - 997 - All Inbound Messages
Known Host Key	Enter a base64-encoded string of your SFTP server. This is used to authenticate your server. SAP Business Network for Logistics can also retrieve the known host key and insert it for you. As a prerequisite, you must have entered the address (including the port) of the SFTP server.
Message Version	The *D96A* message version is selected by default corresponding to EDIFACT. The *V4010* message version is selected by default corresponding to ANSI X12. No version will be displayed for message type *All Inbound Messages*.
Description	The *Instructions of Transport* description is selected by default corresponding to EDIFACT message type IFTMIN. The *Load Tender* description is selected by default corresponding to ANSI X12 message type 204. The description will not be shown for message type *All Inbound Messages*.
Directory	Enter the relative path to write the file to a directory. Example: `parentdirectory/childdirectory`
File Name	Enter the name of the file to be written. The entry is valid only for LBN outbound message types. Additional allowed placeholders: - `%DOCUMENT_ID%` - `%MESSAGE_TYPE%` - `%TIMESTAMP%` - `%SHIPPER_LBN_ID%` - `%CONTROL_NUMBER%` The placeholder `%CONTROL_NUMBER%` is only available for ANSI X12 997 messages.

Attribute	Description
	You can combine these placeholders with any fixed strings and in any order (`<string 1><placeholder 1><string 2><placeholder 2><string 3>`). Examples: • `LBN_%MESSAGE_TYPE%_%DOCUMENT_ID%_%TIMESTAMP%.edi` • `%TIMESTAMP%_%SHIPPER_LBN_ID%.edi`

When creating Basic authentication details, the provided user name and password are used for authentication during the API call to your system.

When creating API Key authentication details, you need to enter the following information:

Attribute	Description
Name	An identifier for the details
API Key Attribute	Enter the key for the request header.
API Key Value	Enter the token to be used for authentication.

The API key represents the name of the token that a client provides when making API calls. The token is sent as request header.

When creating OAuth 2.0 authentication details, you need to enter the following information:

Attribute	Description
Name	Enter a name to identify the authentication details in this app.
Grant Type	Valid values are: **Client Credentials** (default) and **Client Certificate**.
Authentication URL	Enter the URL of the OAuth 2.0 authorization server that issues the access token.
Client ID	Enter the ID of the client that you're connecting to.
Client Secret	Secret key of the client that you're connecting to. OAuth 2.0 uses a multiple-step authentication pattern: Client credentials (**Client ID** and **Client Secret**) are used by the client application to initially request an access token. The access token is then used to authorize the client (for as long as the token is valid) to access the server's resources (for example, the resources that are used in the associated integration flow). In many OAuth 2.0 scenarios, the access token is issued (or generated) by an authorization server.
Client Authentication	Select **Send as Body Parameter** or **Send as Request Header**. This enables access to your system. If you select **Send as Body Parameter**, the **Client ID** and **Client Secret** are sent in the request body to the authentication server.

Attribute	Description
	If you select **Send as Request Header**, the **Client ID** and **Client Secret** are sent as a part of the authorization header to the authentication server. Valid values are: client credentials and client certificate.
Client Certificate	When you select **Send as Body Parameter**, the Client ID goes in the request body to the authentication server. If you choose **Send as Request Header**, the Client ID becomes part of the authorization header sent to the authentication server. Specify the path of the client certificate (key pair file) in the **Key Pair File** field. Enter its password in the **Password** field.
Include Scope	Select the checkbox to provide the OAuth 2.0 scope information that will be included in the request body.
Scope	Enter the scope that you defined for the endpoint.
Content Type	`application/x-www-form-urlencoded` is selected by default.

Assigning Authentication Details and a Base URL to a Service

Prerequisites

You have specified the communication details (base URL) and authentication details (basic or OAuth 2.0 credentials, or API key).

Context

Procedure

1. In the **System Connections** app, on the **Outbound Connections** view, choose **Create** and complete the **Create Service** dialog. The table below provides more information.

Field	Explanation	Usage
Service Name	This is the list of services supported by SAP Business Network for Logistics for communicating with an external system. These services are groupings of APIs for a specific scenario such as freight tendering, freight subcontracting and freight tracking.	SAP Business Network for Logistics supports the following services: • Resource Road Tracking FTL (Removed) • Shipment Road Tracking LTL • Shipment Tracking All • Shipment Road Tracking FTL • Resource Road Tracking LTL (Removed) • Shipment Ocean Tracking FCL

Field	Explanation	Usage
		• Resource Ocean Tracking FCL (Removed) • Shipment Ocean Tracking LCL • Resource Ocean Tracking LCL (Removed) • Resource Tracking All (Removed) • Freight Tendering All • Freight Subcontracting All • Shipment Air Tracking All • Supply Chain Risk • Freight Ocean Booking All • Freight Air Booking All • Freight Road Subcontracting FTL • Freight Road Subcontracting LTL • Freight Road Tendering FTL • Freight Road Tendering LTL • Resource Tracking Parcel (Removed) • Shipment Tracking Parcel • Shipment Rail Tracking All • EDI Inbound All For more information, see Automating EDI Pulling. • Acknowledgement • Freight External Planning All **⚠ Caution** Resource tracking related services have been removed on August 17th, 2024. You can use the shipment tracking related services instead.
Operation Name	Based on the selected service, you can select an operation name from the list. These operation names are the APIs that SAP Business Network for Logistics uses to call an external system.	The following operations are supported for freight tendering scenarios: • `Post freight request for quotation` • `Post freight request for quotation cancellation` • `Post freight quotation notification`

Field	Explanation	Usage
		The following operations are supported for freight subcontracting scenarios: • `Post freight order for subcontracting` • `Put freight order for subcontracting` • `Post freight order for subcontracting cancellation` • `Post freight order for subcontracting notification` The following operations are supported for freight booking (ocean) subcontracting scenarios: • `Post freight booking request` • `Post freight booking request cancellation` • `Post freight booking shipping instruction` The following operations are supported for freight booking (air) subcontracting scenarios: • `Post freight booking request` • `Post freight booking request cancellation` The following operations are supported for freight tracking scenarios: • `Put order for tracking` • `Delete order for tracking` The following operations are supported for supply chain risk scenarios: • `Put location data` • `Delete location data` The following operations are supported for EDI scenarios: • `Pull EDI` • `Post Acknowledgement` The following operations are supported for external planning scenarios:

Field	Explanation	Usage
		• `Post Freight Document Generic Request` • `Delete Freight Document Generic Request`
Message Format	Defines the communication format for external communication.	The following values are supported: • API • EDI • IFTSTA
Communication Details	Defines the endpoint that SAP Business Network for Logistics calls.	The communication details are fetched as maintained in the **Manage Details** view. You are recommended to use different communication details for each API operation. EDI format can be sent to SFTP and email. API calls can be made to HTTPS endpoints only.
Authentication Details	Defines the authentication that SAP Business Network for Logistics uses when calling an external system. If the communication detail is **Email**, the authentication details are not required.	The authentication details are fetched as maintained in the **Manage Details** view.
Activate the Operation	Defines if SAP Business Network for Logistics uses the connection.	Activate or deactivate

2. To test the service, select it in the table and choose one of the two methods from the **Test Service** drop down.
    - With Default Payload

      An internal shipper tenant is used to trigger the payload. The payload content is static with the intention of testing the connection from SAP Business Network for Logistics to your carrier system.
    - With Custom Payload

      An internal shipper tenant is used to trigger the payload. You can modify the payload content to the extent of changing the order details with the intention of testing various scenarios. However, the order ID and the shipper LBN ID will always be fixed by SAP Business Network for Logistics. The test service will not work with any other shipper tenant.

      You will find a sample payload shippers would send from their system here .

The service test is only available for test tenant types for the following operations:

- `Post freight request for quotation`
- `Post freight request for quotation cancellation`
- `Post freight quotation notification`
- `Post freight order for subcontracting`

- Post freight order for subcontracting cancellation
- Post freight booking request
- Freight booking shipping instruction
- Delete order for tracking
- Post freight booking request cancellation
- Put order for tracking
- Put freight order for subcontracting
- Post freight order for subcontracting
- Post freight document generic request
- Put location data
- Delete location data

Automating EDI Pulling

This feature allows you to schedule automated EDI pulling operations for ANSI and EDIFACT inbound messages. You can run the pulling job during a specific time period each day or on specific days of the week.

Prerequisites

Ensure you set the **Service Name** to **EDI Inbound All** and the **Operation Name** to **Pull EDI** in the **Create Service** dialog when using the **System Connections** app to create a connection service.

Procedure

1. Specify the recurrence type of the EDI pulling service by using the **Recurrence** toggle.
    - **Daily** (default): The service runs every day.
    - **Weekly**/**On Days**: The service runs on selected days of the week.
2. Set the time period during which the service runs based on the selected recurrence type by specifying the **Recur Start Time** and **Recur End Time**.
3. Choose the interval for the service from the **Interval** drop-down list.
4. Choose your time zone from the **Timezone** drop-down list. This setting applies to the time period you set in step two.

Enabling Inbound Communication

Context

To call the provider APIs published by SAP Business Network Global Track and Trace, complete the following steps.

In the **System Connections** app under **Connection Settings**, you can enable inbound connections for **Multi-Cloud** and generate OAuth 2.0 authentication credentials on your own without creating a support ticket.